

Exercise 1: SQL Fundamentals

Database: employee_db

id	first-name	last-name	department	salary	hire-date	city
1	John	Doe	IT		2018	NY
2	Jane	Smith	HR		2019	Ch
3	Mike	Johnson	Finance		2017	LA
4	Carol	Brown	IT			NY
5	David	White	Marketing			SF
6	Emily	Davis	IT			C
7	Robert	Wilson	Finance			H
8	Jessica	Moore	HR			LA
9	Daniel	Claik	Marketing			C
10	Laura	Hall	IT			SA

1. SELECT statement

Write a SQL query to retrieve all columns from the employee table.

```
SELECT *
```

```
FROM employees;
```

2. SELECT DISTINCT statement

Write an SQL query to find all the unique departments in the employee table.

```
SELECT DISTINCT departments  
FROM employees;
```

department
IT
HR
Finance
Marketing

③ ORDER BY Statement

Write an SQL query to retrieve all employees first and last names, ordered by salary in descending order.

```
SELECT first-name, last-name, salary
FROM employees
```

```
ORDER BY salary DESC;
```

First-name	Last-name	Salary
① Emily	Davis	62000
② Mike	Johnson	60000
③ Robert	Wilson	59000
④ John	Doe	55000
⑤ Sarah	Brown	53000
⑥ Daniel	Clark	52000
⑦ David	White	52000
⑧ Teresa	Moore	51000
⑨ Laura	Hall	50000
⑩ Jane	Smith	48000

④ Limit Statement

Write an SQL query to retrieve the top 5 highest paid employees.

```
SELECT first-name, last-name, salary  
FROM employees
```

```
ORDER BY SALARY DESC;
```

LIMIT 5;

first-name	last-name	salary
Emily	Davis	62 000
Mike	Johnson	60 000
Robert	Wilson	59 000
John	Doe	55 000
Sarah	Brown	53 000

⑤ WHERE Statement

Write an SQL query to find employees who work in the IT department.

```
SELECT first-name, last-name, department  
FROM employees
```

```
WHERE department = 'IT';
```

first-name	last-name	department
John	Doe	IT
Sarah	Brown	IT
Emily	Davis	IT
Laura	Hall	IT

⑥ AND Statement

Write a SQL query to find employees who work in the finance department AND have a salary greater than 58 000.

```
SELECT first-name, last-name, department, salary  
FROM employees
```

```
WHERE department = 'finance' AND salary > 58000
```

first-name	last-name	department	salary
Julie	Johnson	finance	60 000
Robert	Wilson	finance	59 000

⑦ OR statement

Write an SQL query to find employees who work in the HR department or the marketing department.

```
SELECT first-name, last-name, department  
FROM employees
```

```
WHERE department = 'HR' OR department = 'marketing'
```

first-name	last-name	department
Jane	Smith	HR
David	White	Marketing
Jessica	Moore	HR
Daniel	Clark	Marketing

⑧ NOT statement

Write an SQL query to find employees who do not work in the IT department

```
SELECT first-name, last-name, department  
FROM employees  
WHERE NOT department = 'IT';
```

first-name	last-name	department
Jane	Smith	HR
Mike	Johnson	Finance
David	White	Marketing
Robert	Wilson	Finance
Jessica	Moore	HR
Daniel	Clark	Marketing

⑨ IN statement

Write an SQL query to find employees who are in HR, IT or finance departments.

```
SELECT first-name, last-name, department  
FROM employees
```

```
WHERE department IN ('HR', 'IT', 'finance')
```

first-name	last-name	department
John	Doe	IT
Jane	Smith	HR
Robert	Wilson	Finance

⑩ Combining Conditions

Write an SQL query to find employees who are in the IT department, have a salary greater than 50 000 and are located in New York.

```
SELECT first-name, last-name, department,  
       salary, city
```

```
FROM employees.
```

```
WHERE department = 'IT'
```

```
AND salary > 50 000
```

```
AND city = 'New York'
```

First-name	Last-name	department	salary	city
John	Doe	IT	55 000	NY
Sarah	Brown	IT	53 000	NY

⑪ Combining WHERE, AND and ORDER BY

Write an SQL query to retrieve the first and last names of employees who work in finance or marketing department, earn more than 52 000 and order the result by salary in descending order.

```
SELECT first-name, last-name, department, salary  
FROM employees
```

```
WHERE department = 'finance' OR department = 'marketing'
```

```
AND salary > 52 000
```

```
ORDER BY salary DESC;
```

first-name	last-name	department	Salary
Mike	Johnson	Finance	60 000
Robert	Wilson	Finance	59 000
Daniel	Clark	Marketing	52 000

⑫ Combining SELECT distinct, WHERE and IN
 Write an SQL query to find all the unique cities
 where employees work, excluding those in the
 IT and HR departments.

~~SELECT first-name, last-name, city, department~~

SELECT DISTINCT city

FROM employees

WHERE department NOT IN ('IT', 'HR');

13 Combining WHERE, NOT, AND and ORDER BY

Write an SQL query to retrieve employees who are NOT in the finance department, have salary greater than 50 000 and order the results in hire date in ascending order.

```
SELECT first-name, last-name, department, salary,  
       hire date
```

```
FROM employees
```

```
WHERE department NOT = 'finance'
```

```
AND salary > 50 000
```

```
ORDER BY HIRE_DATE ASC;
```

⑩ COMBINING WHERE, OR, IN AND LIMIT

Write an SQL query to find the first 3 employees who work either Chicago or Los Angeles and belong to the IT or marketing department.

```
SELECT first-name, last-name, department, city  
FROM employees
```

```
WHERE city IN ('Chicago', 'Los Angeles')
```

```
AND department IN ('IT', 'Marketing').
```

```
LIMIT 3;
```